2Team No.: 1 Names: Thaddeus Harris & Elizabeth Squires

Submission 5 – Analysis and Results – Rubric/Checklist & Grade Sheet

Item	Score		Comments
CONTENT			• **Table and figure are mentioned but not properly introduced or explained** • **No page numbers** • **Change "three point" to "three-point" in first and third paragraphs**
Presents the outcome of the experiment, including raw data and calculated results (e.g., stress) (yes(2)/no(0))	2		
Presents observations made during the experiment (yes(2)/no(0))	2		
Explains the analysis process followed through the experiment (yes(2)/no(0))	2		
Includes all figures and tables that are needed to effectively communicate the relationships between measured properties (yes(2)/no(0))	2		
Figures and tables are preceded by a stand-alone explanation of what is depicted in the figure or table, with a reference to the figure or table following the text (yes(2)/no(0))	0		
Results are meaningful toward the goal of the project (evaluating the hypothesis) (yes(2)/no(0))	2		
Written in prose that is connected and flows (e.g., not a series of disconnected paragraphs) (yes(2)/no(0))	2		
WRITING			
Each paragraph has a topic sentence (yes(2)/no(0))	2		
Each sentence after the topic sentence (1 & 2 below):			
(1) Relates to the topic sentence (yes(2)/no(0))	2		
(2) Is well-written (yes(2)/no(0))	2		
Sentences are complete (yes(2)/no(0))	2		
No spelling errors (yes(1)/no(0))	1		
No typographical errors (yes(1)/no(0))	1		
The text effectively communicates ideas and information to both an expert and non-expert audience (yes(1)/no(0))	1		
Writing effectively transitions between ideas/topics (yes(1)/no(0))	1		
FORMATTING			
Margins are 1" and text is double spaced (yes(1)/no(0))	1		
Title page & page numbers included (yes(1)/no(0))	0		
Fonts and text justification are consistent; formatting of paragraphs and section headings/subheadings is consistent throughout the document (yes(1)/no(0))	1		
References are appropriately used and cited in-text when used; references list included at the end of the body of the submission in the correct format (i.e., complete reference information for external sources) (yes(1)/no(0))	1		
Correct word count (400-600 words) (yes(1)/no(0))	1		
TOTAL	**28/31**	**/31**	

Measuring Effects of Different Galvanic Anodes on the Corrosion of Steel

Elizabeth Squires, Thaddeus Y. Harris

Department of Civil and Environmental Engineering, Northeastern University

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Craig M. Shillaber

Analysis and Results

This study has three sources of data. Firstly, the qualitative data taken of the samples before and after in the form of photos to observe any surface corrosion on each of the samples, and to observe loss of material and surface changes in each of the anodes. The second source of data is the weights of the anodes before and after the soaking period. The third and final source of data is the load-deformation data obtained from the three-point bending test on the ADMET machine.

Starting with the qualitative data, each sample experienced different levels of surface corrosion with the anodes experincing either staining on exposed areas, surface corrosion, or spalling.

Figure 1: Photo of zinc anode and associated steel sample

Figure 2: Photo of aluminum anode with associated steel sample

Figure 3: Photo of magnesium anode with associated steel sample

Each of the samples and anodes showed a different pattern of corrosion. The zinc anode was stained in multiple areas, though there was no visually noticeable loss of mass from the start of the expirement. The steel attached to the zinc experienced similar staining with a slightly different

coloration to the anode. The aluminum anode experienced staining where it was in contact with the water (not covered by tape), with the associated sample experiencing relatively even discoloration across the surface, with some streaks of green discoloration at the surface. The magnesium anode experienced the most corrosion out of the three, exhibiting some spalling bahavior. The sample experienced some discoloration and roughness where it contacted the magnesium anode.

The weight changes for each of the galvanic anodes are presented in the following table, with the initial weight being at an air dry state, and the 1-week weight being the weight of the anodes after drying with a paper towel .

Weights (g)	Zinc	Aluminum	Magnesium
Initial	134.8	34.6	39.4
1 Week	134.7	34.6	34
% Change	0.0742	0	13.7056

Table 1: Weight Changes in Galvanic Anodes

The magnesium experienced the largest change in weight over the one week period with the aluminum anode experiencing no change and the zinc anode experiencing a 0.1 gram decrease in weight, though when considering additional moisture the metal may have taken on in pores on the sample this decrease may be more than was observed in this testing. The same is true of the aluminum sample. This is a flaw with the methodology and if this study was to be repeated the samples would be left in the air for a period of time to reduce the amount of water in the samples to an air dry state.

The following graph shows the stress-strain relationships for each of the steel samples, as labeled by the anode they were attached to, or labeled as control in the case of the control sample.

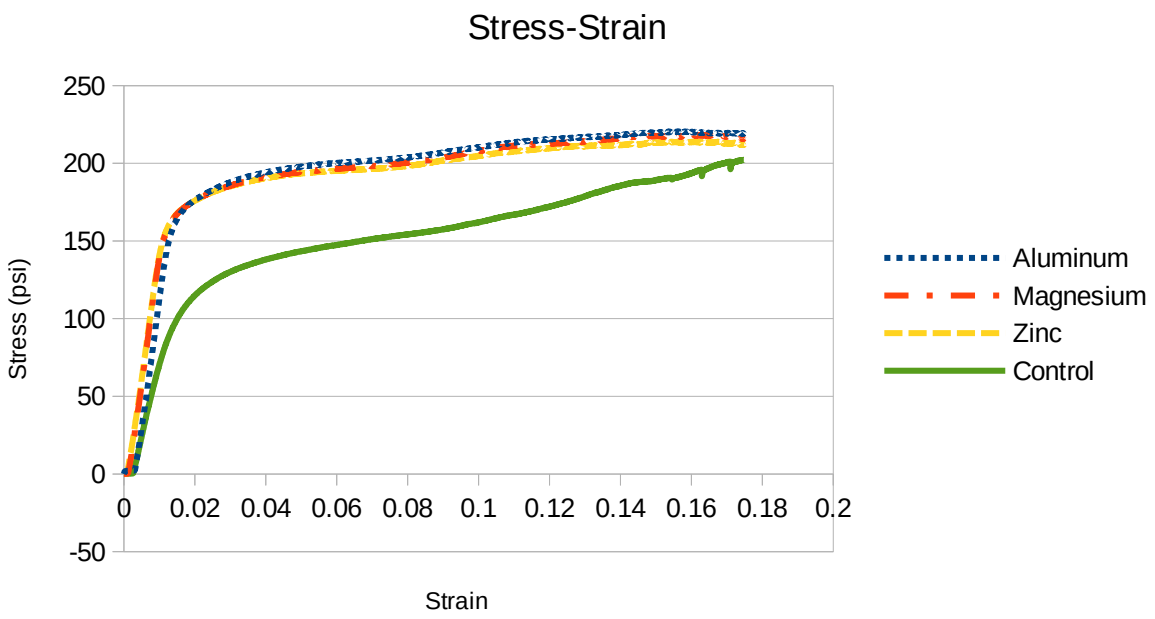

Figure 1: Stress-Strain graph of all samples

Each of the samples attached to a galvanic anode performed about the same, with the sample performing worse initially, having a significantly smaller elastic strength than the treated samples, though any tangent modulus taken after the elastic limit would be larger for the control than it is for the treated samples at the same point on the strain axis. Due to this the stress needed to deform the material approached that of the treated samples as the test progressed. None of the samples experienced fracture during testing, so no ultimate strength values could be observed, and the testing was halted after the steel section was no longer in contact with the top of the supports. These results could provide evidence that the usage of a galvanic anode in saltwater increases the elastic strength of steel and increases the toughness. Alternatively, the difference from the control sample could be due to parameters of the ADMET machine being changed between the time when the control sample was tested and when the treated samples were tested, such as an altered rate of displacement. It was assumed that those parameters would not be alerted by another user between the time when both samples were tested, with

the only parameter that was closely monitored being the distance between the supports on the three-point bending mechanism, at 6cm. This demonstrates a problem in the experimental design and in future testing all parameters of the ADMET machine that are capable of being changed should be specified in the experimental design.

Methods and Theory

The two main qualities being studied in this project are the change in mass of galvanic anodes and the stress-strain graph from a three point bending test for each of the samples. These stress strain relationships are then used to find the elastic strength, Young's modulus, and toughness of each sample. These material properties allow one to determine the suitability of the steel for applications like bridges where steel members are commonly used to take bending loads.

Multiple assumptions were made during the design of this study. The first assumption that was made was that galvanic corrosion could be accelerated by increasing the salinity of the water without compromising the test results by affecting the galvanic anodes in steel in an unintended manner. This assumpton was made as none of the anodes' elements interact with water molecules, and the concentrations of the pieces removed from the anodes by corrosion would be relatively small in the water, presumably not effecting reaction significantly. Another assumption which was made was that the corrosion of the anodes in a stagnant pool of water would not effect the results as compared to a flowing body of water. The three point bending test was chosen under the assumption that it provides a reasonable analogue to the bending stresses that steel undergoes while reinforcing a concrete structure.

To find the change in mass the initial mass had the final mass subtracted from it. This was used to find the percent change in mass by dividing the change in mass by the initial mass and multiplying that number by 100%.

$$\frac{m_0 - m_t}{m_0} \times 100\% = \%\Delta m$$

Equation 1: Percent Change of Mass

The Young's modulus is found by sectioning off the initial linear part of each sample's stress strain graph, and finding the slope of this linear section. While there is a Young's modulus at all points on the stress strain graph, beyond the elastic linear section this number is difficult to calculate as there

is no easily obtainable function to find strain from stress. The relationship between stress (σ) and strain (ε) resulting in the initial modulus of elasticity (E) is:

$$E = \frac{\sigma}{\varepsilon}$$

Equation 2: Young's Modulus

The elastic strength is obtained by finding the amount of strain it takes until the stress-strain graph ceases to follow the initial Young's modulus, which indicates that the material will not return to its original length after the load has been removed. This is found visually from the stress strain graph where the linear section becomes non linear. Elastic strength has no general formula, as it is obtained from collected data, and is not exact, as there is no amount of stress where when it is removed the material will return to the exact initial state, with the elastic strength being qualitative to a degree.

The toughness is obtained by taking the area under the stress strain graph, and dividing it by the volume of the sample. Toughness has no standard equation, as the stress-strain function is not easily obtainable and therefore cannot be integrated to find the area, with the exception of the linear elastic area of the stress strain graph. In addition to this, none of the steel samples experienced fracture, so the elastic toughness is the only form of toughness obtained for this study. As the area under the curve for the linear elastic section of the graph is a triangle the area can be approximated as:

$$\frac{energy}{volume} = \frac{1}{2} \times \sigma\varepsilon$$

Equation 3: Toughness